

Finger Hut – HW5

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Task 3 - Time series # orders shipped

- Important observations from aggregation process:
 - the data doesn't include any ongoing journeys at its start point $\Rightarrow$ there aren't any orders shipped until a few days in. **Solution:** find 80th percentile of successful journey length and start train data there to accurately capture a reflective trend
 - Even though the warehouse is closed on weekends, the data shows orders shipped on quite a few weekends (at a much lower count than weekdays). For daily aggregation, this would show a misleading weekly seasonality effect. **Solution:** (*probably not the best solution!*) drop all weekends from the dataset, the Prophet model interpolates missing dates.
- Without extra information, I would probably use weekly aggregation because of the issue with orders shipped on weekends. It is still reflective of granular seasonality, specifically holidays, which have the biggest impact on orders shipped.
- The most useful additional regressor to use would be seasonal promotions (we don't have that AFAIK).

Plots:

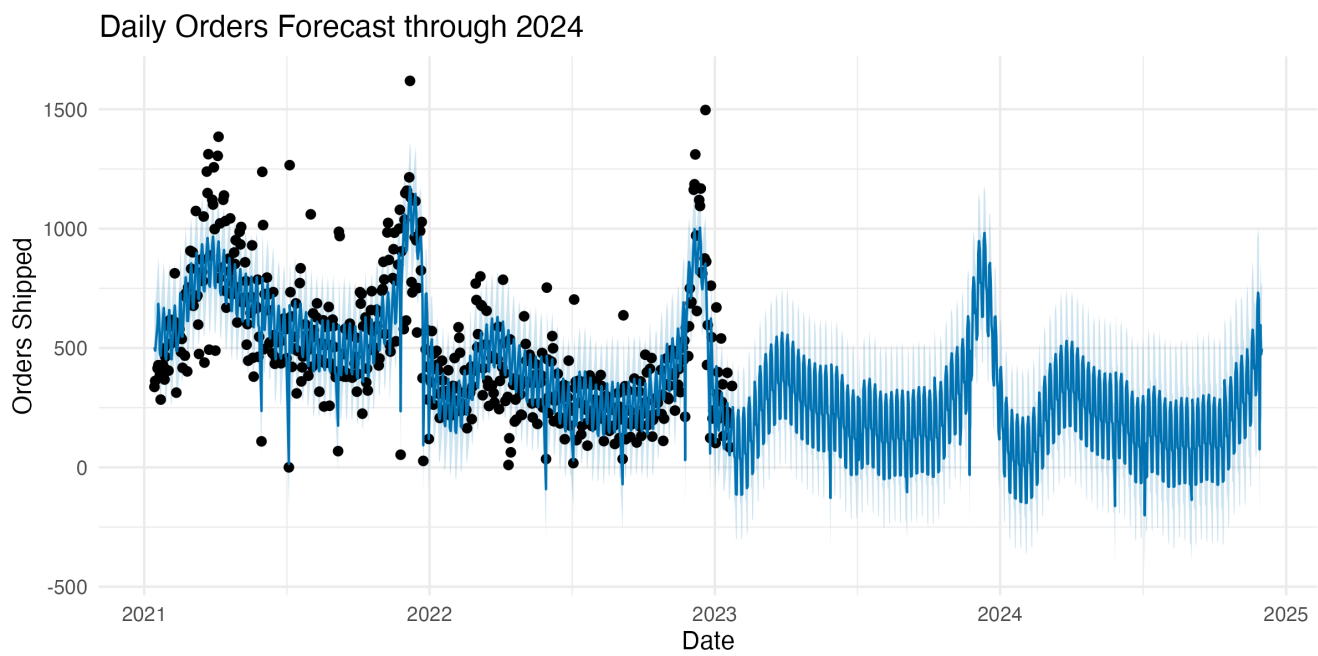


Figure 1: Prophet Daily Forecast plot

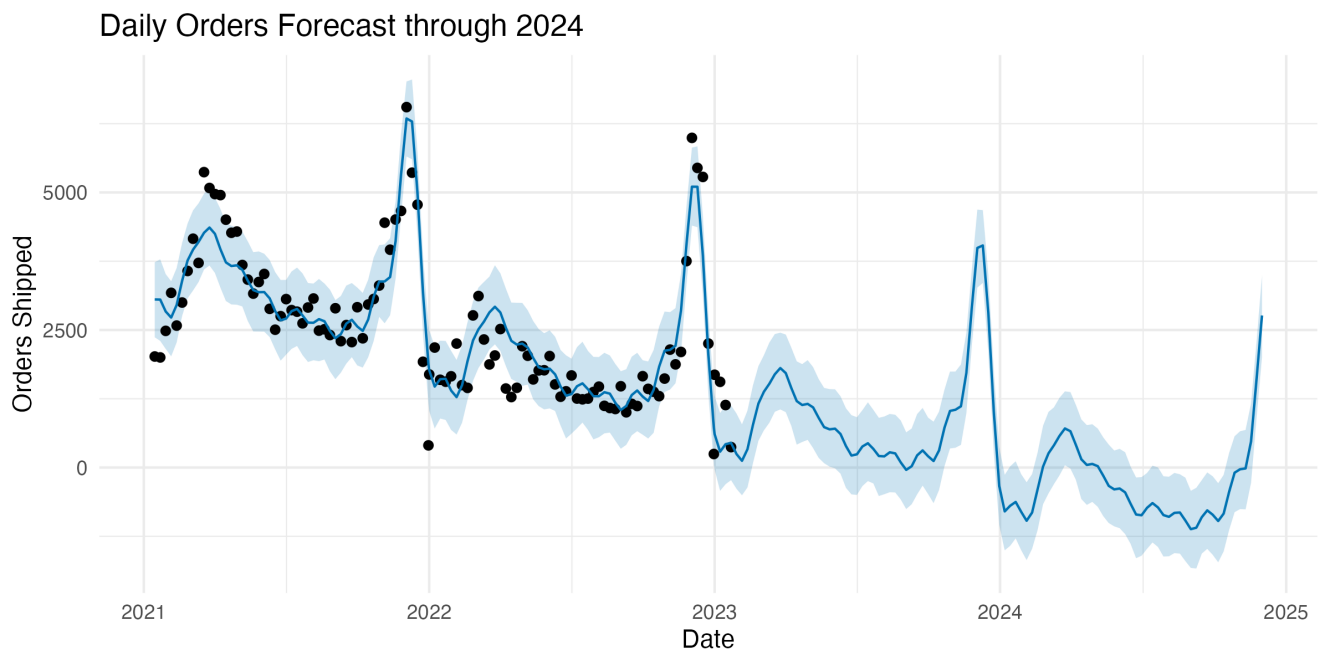


Figure 2: Prophet Weekly Forecast plot

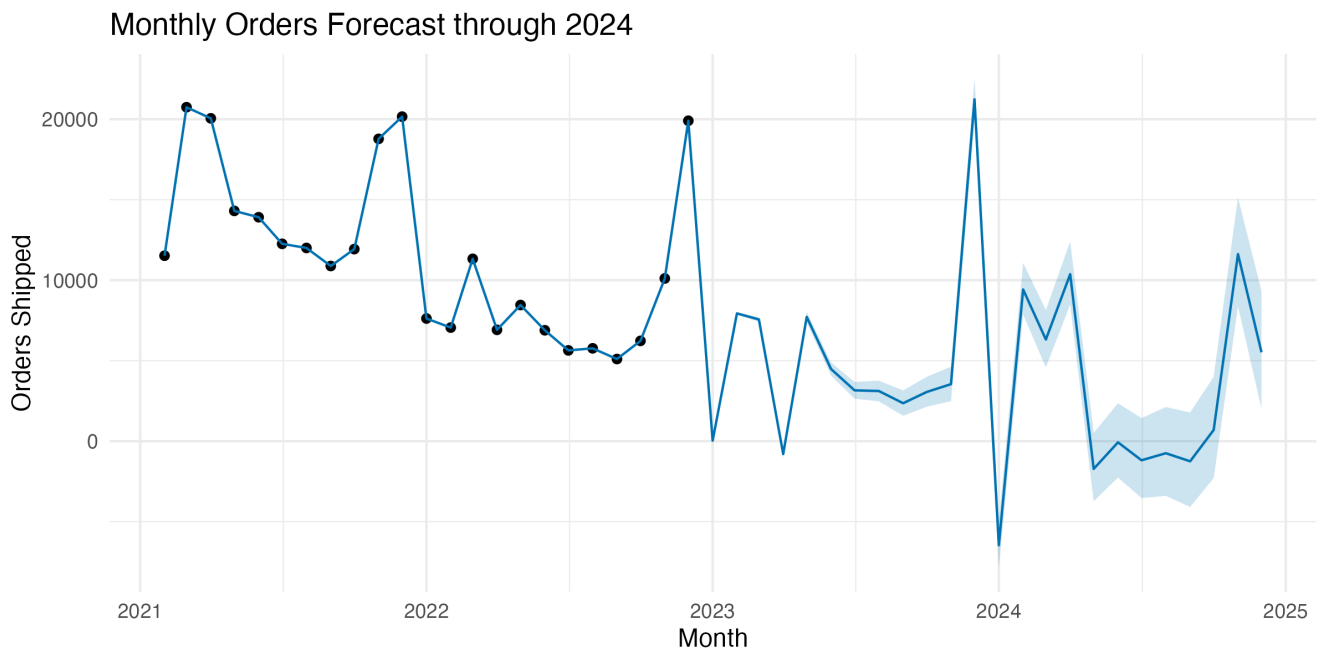


Figure 3: Prophet Monthly Forecast plot

Task 4

Research Question :

- What are the features differentiate **successful** / **Unsuccessful** / **Ongoing Journeys** + general observations

Definitions :

- **Successful Journey:** A journey that eventually ends with an `order_shipped` event. Regardless of how long the journey takes, if an order is shipped, the journey is considered successful.
- **Unsuccessful Journey:** A journey that does not end with an `order_shipped` event and has had no activity for at least 60 days before the cutoff date.
- **Ongoing (Incomplete) Journey:** A journey that has not ended with an `order_shipped` event, but whose most recent activity occurred within 60 days of the cutoff date, meaning the final outcome is still unknown.

Report :

first to last action

- Despite the typical user being on the site for 134 days, **successful** users tend to be on the platform for much less time (48 days) – **Unsuccessful** for 170 days.

- Meaning Journey duration varies substantially by outcome type.

Time Between Actions

- **successful** journeys have the shortest average time between actions (about 0.9 days on **avr**), suggesting that users who eventually purchase tend to move through the process relatively quickly.
 - **successful** journeys **median** Time Between Actions is only 90 seconds (1min, 30 sec), indicating dramatic skew
 - Indicating **successful** journeys typically make quick successive actions
 - Variability : **successful** journeys have a **SD** of 6 day – indicating a wide variability in user enagagment
- **Unsuccessful** journeys have much longer gaps between actions (about 5.6 days on **avr**), suggesting that disengaged users interact far less frequently over time.
 - **Unsuccessful** journeys **median** Time Between Actions is only 120 seconds (2min), indicating dramatic skew
 - Indicating **Unsuccessful** journeys typically make slower but still, successive actions
 - Variability : **Unsuccessful** journeys have a **SD** of 17 day – indicating dramatically more variability than **successful** journeys – suggesting that **Unsuccessful** journeys are far more likely to have longer periods of time of being off the website for longer periods of time. Suggesting those whom are **successful** journeys stay more consistently engaged

Stage Counts

- Across all journey types, the most common stage is **First Purchase**, indicating that most user activity centers around browsing, cart activity, and checkout behavior.
- **successful** journeys show substantially more activity in the **Downpayment** (533,281 > 24,692) and **Credit Account** stages, suggesting these are strong signals of eventual conversion.
- **Unsuccessful** journeys have a much larger proportion of **NA** and **Apply for Credit** events, suggesting many users disengage before reaching later purchasing stages.
- The **Order Shipped** stage appears only in **successful** journeys, as expected, and represents the final conversion event.
 - Sanity Check[^]

Final Stage

- Nearly all **successful** journeys end in the **Order Shipped** stage (about 99.5%), which confirms that successful users progress fully through the purchase funnel.
 - Sanity Check[^]
- Ongoing journeys most commonly end in the **First Purchase** stage (about 47.6%), suggesting many active users are still browsing, shopping, or checking out.
- **Unsuccessful** journeys overwhelmingly end on events with missing (NA) stage labels (about 72%), indicating that many users disengage during promotional or uncategorized interactions before progressing further.
- Among categorized stages, **Unsuccessful** journeys most often stop during the **First Purchase** and **Apply for Credit** stages, suggesting that many users drop out before completing payment or shipment-related steps.
- Very few **Unsuccessful** or **Ongoing** journeys reach the **Downpayment** or **Credit Account** stages, implying that progressing deeper into the funnel is strongly associated with eventual success.

Path Analysis using Stages

- **successful** journeys tend to follow structured progression paths that move through **Apply for Credit**, **First Purchase**, **Credit Account**, and **Downpayment** before ultimately reaching **Order Shipped**.
- Both **Ongoing** and **Unsuccessful** journeys are heavily concentrated in early funnel stages, especially **Apply for Credit** and **First Purchase**, suggesting many users never progress deeper into the purchasing pipeline & **Unsuccessful** users are those whom are repeatedly rejected after applying for credit.
- Repeated sequences such as long chains of **Apply for Credit** indicate that many users repeatedly interact with credit-related activities without converting.
- The presence of **Discover** stages is more common among **Ongoing** and **Unsuccessful** journeys, suggesting that many users remain in exploratory or marketing-related behaviors rather than advancing toward purchase completion.

Down payment Stage

- Nearly all **successful** journeys (99.3%) reached the **Downpayment** stage, suggesting that progressing to downpayment activity is one of the strongest indicators of eventual order shipment success.
- Very few **Ongoing** (2.4%) or **Unsuccessful** (1.7%) journeys reached the Downpayment stage, indicating that most users who fail to convert never progress that deeply into the purchasing funnel.
- Downpayment stage appears to strongly separate successful journeys from non-successful journeys and is likely a highly predictive feature for future models.